# MP$^3$: Mining Pure Pivot Patches for Robust Real-World Single Image Defocus Deblurring

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***Abstract*—Single Image Defocus Deblurring (SIDD) aims to restore an all-in-focus image from a partially defocused image. Early methods resort to designing complex dynamic blocks to enhance spatial awareness but their generalization performance is limited. Recently, prior-driven approaches leverage the defocus kernel prior to achieve promising results but still rely heavily on heuristic priors. In this paper, we rethink how to train a model that generalizes well from a data-quality perspective. Specifically, we find that not all patches contribute equally during training and aim to Mine Pure Pivot Patches (*MP*$^3$) that offer rich and clean learning signals for training a robust and generalizable model. To achieve this, we design a metric that quantifies the quality of patches by measuring the entropy-density decay and residual sparsity. Entropy-density decay is aimed at evaluating the richness of defocus cues (*related to Pivot*) while residual sparsity is for evaluating whether a patch pair is aligned with the defocus degradation (*related to Purity*). Additionally, a simple yet effective scale-recurrent neural network is adapted from existing methods by removing redundant activations and operations while emphasizing cross-scale selections. To further enhance robustness, we design a curriculum frequency learning paradigm to progressively guide the model to learn from structures to details. Through extensive experiments, our method demonstrates strong generalization performance across different datasets.**

***Index Terms*—Defocus deblurring, Patch sampling, Entropy-density decay**

## I. Introduction

Capturing a scene with large depth variation using a wide aperture causes regions near the focal plane to appear sharp while others become blurred, a phenomenon known as defocus blur. The spatially varying nature of this blur makes its removal challenging. This paper addresses the task of Single Image Defocus Deblurring (SIDD), which aims to recover an all-in-focus image from a single defocused input.

Existing SIDD methods can be categorized into three types. *(i)* Two-stage conventional methods (*e.g.* [5]–[10]): They adopt a two-stage paradigm that first estimates defocus kernels and then performs deconvolution. This framework is interpretable and simple but prone to error propagation. *(ii)* Early end-to-end learning methods (*e.g.* [2], [3], [11]–[14]): They directly learn a mapping from defocused to sharp images via complex dynamic filtering blocks or auxiliary tasks. However, this practice often overlooks inherent defocus priors, leading to inferior generalization across datasets. *(iii)* Recent prior-driven end-to-end methods (*e.g.* [1], [15]–[21]): They embed defocus priors into neural network design, imposing explicit or implicit regularization and achieving promising generalization.

Despite their success, most prior-driven methods remain constrained by heuristic designs focused mainly on kernel appearance priors. In contrast, we revisit the problem from a data-quality perspective. Given a defocused image, pixel recovery difficulty depends on depth difference from the focal plane (i.e., blur level). This raises a natural question: *Are objects at the same depth equally easy to deblur?* Empirically, we find this assumption does not hold.

As shown in Fig. 1(a), regions with similar depth produce varying recovery results across SIDD methods. Textured regions yield robust reconstructions with fewer artifacts (pivot examples), while flat regions often suffer from under-deblurring (e.g., GKM [1]) or over-deblurring (e.g., [2], [3]). We therefore hypothesize that training patches can be categorized into *pivot* and *non-pivot* ones. Pivot patches contain rich defocus cues strongly related to both blur level and texture richness, whereas non-pivot patches exhibit weak cues due to minimal blur or flat textures. In this paper, we propose a metric to identify pivot patches for effective training.

Considering only whether a patch pair is pivot is insufficient, as *impure* patch pairs with quality issues may provide strong but misleading learning signals. As illustrated in Fig. 1(b), real-world defocus datasets contain issues such as misalignment, brightness inconsistencies and overexposure that, though rare, can severely confuse the model. In this paper, we propose to detect such impure patches by evaluating their residual sparsity.

Furthermore, purity is intertwined with the pivot property, that is, non-pivot patches are more prone to impurity. In an ideal defocus dataset without quality issues, textured regions under the same blur level provide richer cues for learning, while flat regions, being nearly identical to the sharp labels, lead the model to learn an almost identity mapping without harm. However, in real-world datasets, aperture adjustment introduces brightness inconsistencies. Consequently, flat regions are more affected by such inconsistencies, whereas textured regions remain dominated by defocus cues. Thus, learning signals from flat regions may become detrimental, while those from textured regions stay relatively reliable.

Regarding the Neural Network (NN) architecture, most state-of-the-art methods (*e.g.* [1], [17]–[20]) adopt a scale-recurrent design (e.g., ConvLSTM) that exploits cross-scale

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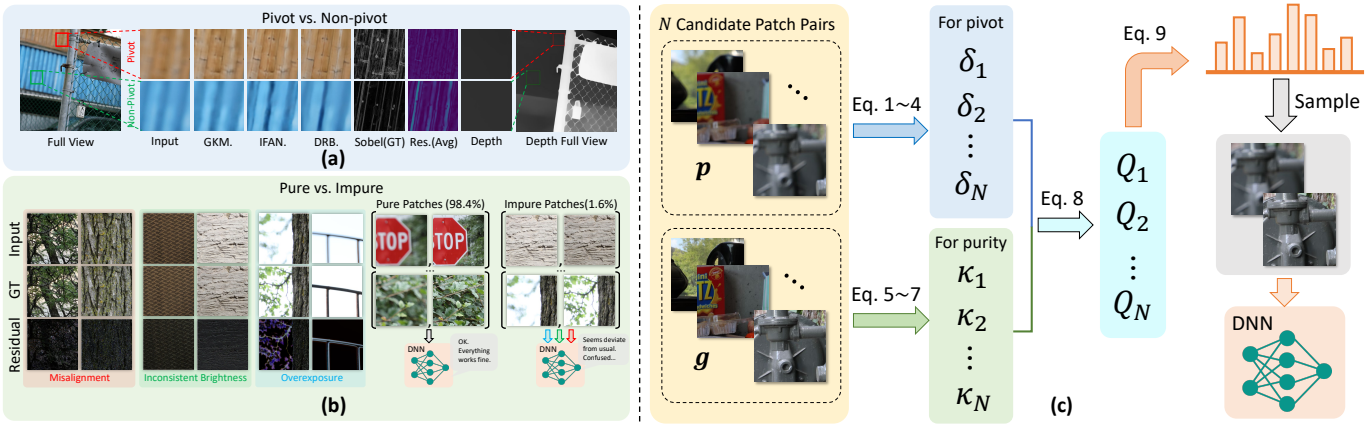



Fig. 1. (a) Deblurring results from different methods. Pivot/Non-pivot represents region with good/bad recovery, respectively. Sobel(GT) represents edge map extracted from GT by the Sobel operator. Averaged residual map on three methods [1]–[3] are visualized to indicate recovery quality. The output depth from DepthAnything [4] is shown to indicate similar defocus blur level. (b) Summary of possible quality issues existing in real-world defocus datasets, such as misalignment, inconsistent brightness and overexposure, which can possibly confuse models. (c) Framework of our proposed MP$^3$.

self-similarity of defocus blur. We simplify this by removing unnecessary activations and operations, proposing a simple yet efficient architecture.

For image deblurring, frequency-domain losses are effective and widely adopted. We introduce a curriculum frequency loss that guides the network to progressively learn from coarse structures to fine details. Unlike prior works that integrate frequency information into NN module design [22], [23], ours emphasizes loss design.

To conclude, our main contributions are summarized as follows:

- We propose a metric to assess the pivotness and purity of training patch pairs, emphasizing samples with rich and clean learning signals.
- We introduce a curriculum frequency loss that guides the NN to progressively learn from coarse structures to fine details.
- We design a simple and efficient scale-recurrent architecture for SIDD.

## II. Related Works

### A. Two-stage Conventional Methods

Most early SIDD methods adopt a two-stage pipeline that first estimates a defocus map or blur kernel (*e.g.* [5]–[10], [24], [25]) and then performs non-blind deconvolution. However, their performance heavily depends on accurate defocus estimation, as small errors can cause global ringing artifacts and degrade quality.

### B. Early End-to-end Learning Methods

Early deep learning methods use end-to-end networks to map defocused images to sharp ones, avoiding error accumulation in two-stage pipelines. Some design dynamic filtering modules (*e.g.* [2], [3]) for spatially adaptive processing, though flexibility risks overfitting. Others employ auxiliary tasks such as predicting defocus maps [11] or dual-pixel views [13], but these require extra datasets and annotations. Diffusion models [26] are also introduced for SIDD.

### C. Recent Prior-driven End-to-end Learning Methods

Prior-driven methods embed defocus priors into network design to improve efficiency and generalization. Early work models blur kernels with Gaussian Kernel Mixture (GKM) and unrolls deblurring via fixed-point iteration [1]. Later approaches relax this via recursive/grouped kernels [17], [19] or model inverse kernels using implicit neural representations [18]. To further enhance robustness, mixed kernel representations [20] and Gaussian-blob-based formulations [21] have been proposed.

### D. Improvement from Data Perspective

Several works [3], [16], [27] address quality issues in real-world defocus datasets, mainly focusing on misalignment. Ruan *et al.* [3] pretrain on aligned synthetic light-field data and finetune with a misalignment-tolerant VGG loss, while Li *et al.* [16] use an optical-flow network for explicit alignment. However, flow estimation is unreliable on defocused images and may distort defocus signals. In contrast, we focus on mining pure pivot patches to enhance robustness and highlight overlooked issues such as brightness inconsistency and overexposure.

## III. Methods

### A. Metric for Evaluating Quality of Patches

Fig. 1(c) illustrates the proposed MP$^3$ framework. Below we introduce details regarding the metric design and patch sampling. The metric is computed based on entropy-density decay and the residual sparsity of the defocus degradation. We denote a cropped input patch as $\boldsymbol{p} \in \mathbb{R}^{h\times w\times c}$ and its corresponding ground truth as $\boldsymbol{g} \in \mathbb{R}^{h\times w\times c}$.

**Entropy-density decay $\delta$.** For efficiency, we adopt a simple procedure to obtain entropy maps $\boldsymbol{E_p}, \boldsymbol{E_g} \in \mathbb{R}^{h\times w}$: each image is first converted to grayscale, after which a Sobel

operator is applied to extract its edge responses. This yields a lightweight approximation of local information density, providing an efficient surrogate for entropy. Then we define entropy-density decay $\delta$ as below,

$$\boldsymbol{M_E} = \mathbb{1}[\boldsymbol{E_p}(x) > \tau_{\boldsymbol{p}}] \vee \mathbb{1}[\boldsymbol{E_g}(x) > \tau_{\boldsymbol{g}}], \tag{1}$$

$$D_{\boldsymbol{p}} = \frac{1}{\|\boldsymbol{M_E}\|_1} \sum_x \boldsymbol{M_E}(x)\, \boldsymbol{E_p}(x), \tag{2}$$

$$D_{\boldsymbol{g}} = \frac{1}{\|\boldsymbol{M_E}\|_1} \sum_x \boldsymbol{M_E}(x)\, \boldsymbol{E_g}(x), \tag{3}$$

$$\delta = D_{\boldsymbol{g}} - D_{\boldsymbol{p}}, \tag{4}$$

where $\boldsymbol{M_E} \in \mathbb{R}^{h \times w}$ is a binary mask that identifies the union support region where either the edge magnitude of $\boldsymbol{p}$ is larger than $\tau_{\boldsymbol{p}}$ or the edge magnitude of $\boldsymbol{g}$ is larger than $\tau_{\boldsymbol{g}}$. $D_{\boldsymbol{p}}$ and $D_{\boldsymbol{g}}$ define the entropy density within the support region. The entropy-density decay $\delta$ is computed following Eq. 4. Intuitively, $\delta$ should be ideally proportional to the richness of learning signals since it considers both the blur level and the complexity of underlying textures.

**Residual sparsity $\kappa$.** Misalignment, inconsistent brightness, and overexposure cannot be faithfully revealed by entropy-density decay $\delta$. To address this, we exploit the residual sparsity prior in defocus degradation. Specifically, the residual map between a defocused image and its sharp counterpart should remain sparse, regardless of the blur level. The residual sparsity $\kappa$ of a patch pair is formulated as follows,

$$\boldsymbol{R}(x) = |\boldsymbol{g}(x) - \boldsymbol{p}(x)|, \tag{5}$$

$$\boldsymbol{M_R}(x) = \mathbb{1}[\boldsymbol{R}(x) > \tau_{\boldsymbol{R}}], \tag{6}$$

$$\kappa = 1 - \frac{1}{\|\boldsymbol{M_R}\|_1} \sum_x \boldsymbol{M_R}(x)\boldsymbol{R}(x), \tag{7}$$

where $\boldsymbol{R}$ is the residual map between $\boldsymbol{p}$ and $\boldsymbol{g}$, and $\boldsymbol{M_R}$ defines the support region for computing sparsity.

Simultaneously low residual sparsity and low entropy-density decay indicate that the patches contain limited defocus cues but strong learning signals, which is very likely non-pivot and impure. Examples can be found in Sec. IV-C.

**Quality metric $Q$.** Given $\delta$ and $\kappa$, the final quality metric $Q$ is defined as combination of these two metrics using balancing factor $\alpha$,

$$Q = \alpha \cdot \delta + (1 - \alpha) \cdot \kappa. \tag{8}$$

Finally, from $N$ uniformly sampled candidate patches $\{(\boldsymbol{p}_i, \boldsymbol{g}_i)\}_{i=1}^N$, we sample a patch pair according to the following probability distribution,

$$prb_i = \frac{(Q_i + \epsilon)^\lambda}{\sum_{j=1}^N (Q_j + \epsilon)^\lambda}, \tag{9}$$

where $\epsilon$ and $\lambda$ are smoothing factors. Through this biased sampling mechanism, we prioritize patches that contain informative and reliable defocus cues. Full pseudo code for sampling can be found in the Supp.

### B. Curriculum Frequency Learning

We divide the training process into $S$ curriculum stages. At each stage $s \in \{1, \ldots, S\}$, we apply a frequency-domain loss that focuses on a progressively wider band of frequencies.

**Curriculum Gaussian frequency mask.** Let $\mathcal{F}(\cdot)$ denote the Fourier transform with shifting and $(u, v)$ be frequency coordinates. For stage $s$, we define a Gaussian mask

$$\boldsymbol{M}_s(u, v) = \exp\left(-\frac{u^2 + v^2}{2\sigma_s^2}\right), \tag{10}$$

where $\sigma_s$ controls the bandwidth of frequencies preserved in stage $s$. We choose $\sigma_1 < \sigma_2 < \cdots < \sigma_S$ so that the model gradually transitions from learning coarse structures to fine details.

**Curriculum masked frequency loss.** Given the prediction $\tilde{\boldsymbol{p}}$ and ground truth $\boldsymbol{g}$, their masked Fourier spectra at stage $s$ are

$$\widehat{\boldsymbol{p}}_s = \frac{1}{\|\boldsymbol{M}_s\|_1} \boldsymbol{M}_s \odot \mathcal{F}(\tilde{\boldsymbol{p}}), \tag{11}$$

$$\widehat{\boldsymbol{g}}_s = \frac{1}{\|\boldsymbol{M}_s\|_1} \boldsymbol{M}_s \odot \mathcal{F}(\boldsymbol{g}). \tag{12}$$

The frequency-domain loss for stage $s$ is defined as the L1 distance between the masked spectra:

$$\mathcal{L}_{\text{freq}}^{(s)} = \|\widehat{\boldsymbol{p}}_s - \widehat{\boldsymbol{g}}_s\|_1 . \tag{13}$$

The final curriculum learning objective at stage $s$ additionally includes MSE loss in spatial domain and LPIPS loss in feature domain following [18]:

$$\mathcal{L}^{(s)} = \lambda_{\text{MSE}}\mathcal{L}_{\text{MSE}} + \lambda_{\text{LPIPS}}\mathcal{L}_{\text{LPIPS}} + \lambda_{\text{freq}}\mathcal{L}_{\text{freq}}^{(s)}, \tag{14}$$

where $\lambda_{(\cdot)}$ are coefficients balancing different loss terms.

### C. DNN Architecture

Inspired by state-of-the-art SIDD methods, we propose a simple yet efficient scale-recurrent architecture that emphasizes cross-scale selection. In practice, defocus blur varies spatially, and larger blur requires a larger receptive field. Naively increasing network depth or kernel size incurs substantial computational cost. A more economical strategy is to handle severe blur at coarse scales and finer blur at higher resolutions. Accordingly, our design focuses on effective cross-scale selection while removing unnecessary activations and operations, resulting in better efficiency than the widely adopted ConvLSTM-based architectures (*e.g.* [1], [17]–[20]).

As shown in Fig. 2, the input image is downsampled into three scales $\{\boldsymbol{p}^{(t)}\}_{t=1}^3$. Each scale is first processed by a weight-sharing UNet, and the resulting features are fed into a Scale-recurrent Cross-scale Selection (SCS) module. Let $f^{(t)}$ denote the UNet feature at scale $t$ and $h^{(t-1)}$ the hidden state from the previous scale. The computation at scale $t$ is then given by:

$$\widehat{f}^{(t)} = \text{AvgPool}_{2\times2}(\text{Conv}(f^{(t)})), \tag{15}$$

$$\mathcal{A}^{(t)} = \text{Sigmoid}(\text{Conv}(h^{(t-1)} \copyright \widehat{f}^{(t)})), \tag{16}$$

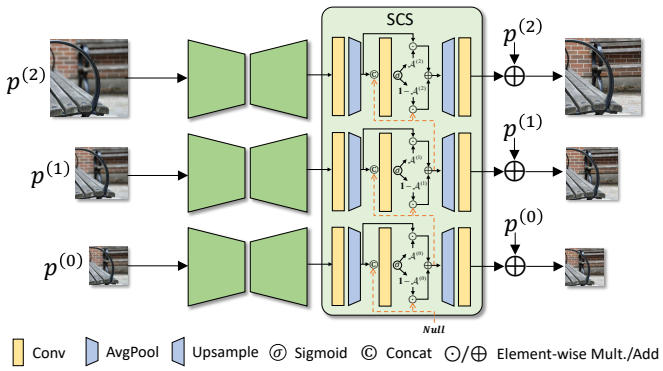



Fig. 2. Proposed scale-recurrent network with cross-scale selection.

$$h^{(t)} = \mathcal{A}^{(t)} \odot \widehat{f}^{(t)} + (\mathbf{1} - \mathcal{A}^{(t)}) \odot h^{(t-1)}, \tag{17}$$

where the attention map $\mathcal{A}^{(t)}$ acts as a cross-scale selector, spatially determining whether the coarser scale provides a better estimation for deblurring specific regions. Then the final prediction $\widehat{\boldsymbol{p}}^{(t)}$ is computed as

$$\widehat{\boldsymbol{p}}^{(t)} = \boldsymbol{p}^{(t)} + \text{Conv}(\text{Up}_{2\times2}(h^{(t)})). \tag{18}$$

During training, the loss $\mathcal{L}^{(s)}$ in Eq. 14 is computed at each scale's output and summed to form the total loss.

## IV. Experiments

### A. Datasets and Implementation Details

**Benchmark datasets.** Following previous works, experiments are conducted on benchmark datasets including DPDD [28], LFODF [11], RealDOF [2], RTF [6], and CUHK-BD [29]. Detailed information on these datasets is provided in the Supp. Among them, CUHK-BD is collected from the Internet without ground truths and is used for qualitative comparison.

**Implementation details.** We train our model using $S = 3$ curriculum stages, each with the same number of iterations, for a total of 500k steps with a batch size of 4. Within each stage, we adopt the Adam optimizer and apply cosine-annealed learning rate decay from $2 \times 10^{-4}$ to $1 \times 10^{-5}$. Data augmentation includes random flipping, rotation, and cropping to $256 \times 256$ patches. Additional details are provided in the Supp. Our Pytorch implementation is available at https://github.com/xinyao240/MP3.

### B. Performance Comparison

**Baselines for comparison.** We compare our method with 15 representative approaches, including DPDNet [28], AIFNet [11], MDP [13], KPAC [15], IFANet [2], GKMNet [1], DRBNet [3], MPRNet [16], Restormer [30], INIKNet [18], NRKNet [17], P2IKT [20], GGKMNet [19], PPTFormer [31], and DefocusGS [21]. We report results from the original publications when available; otherwise, we retrain the models using their released codes. All methods are evaluated at the original input resolution of each benchmark dataset without any downsampling.

**Quantitative results.** Following existing works [17]–[19], [21], we independently train two models on DPDD and LFDOF and evaluate them on their respective test splits. Quantitative results are reported in Tab. I. To assess generalization, both models are further tested on RealDOF and RTF, with results shown in Tab. II. Our method achieves consistently competitive performance across datasets, particularly strong generalization on RealDOF and RTF.

TABLE I
QUANTITATIVE COMPARISONS ON DPDD/LFDOF. BEST AND SECOND BEST ARE HIGHLIGHTED.

| Model | Source | DPDD | | | LFDOF | | |
|---|---|---|---|---|---|---|---|
| | | PSNR | SSIM | LPIPS | PSNR | SSIM | LPIPS |
| DPDNet [28] | ECCV 2020 | 24.348 | 0.747 | 0.277 | - | - | - |
| AIFNet [11] | TCI 2021 | 24.213 | 0.742 | 0.309 | 29.677 | 0.884 | 0.202 |
| IFANet [2] | CVPR 2021 | 25.366 | 0.789 | 0.217 | 29.787 | 0.872 | 0.156 |
| KPAC [15] | ICCV 2021 | 25.221 | 0.774 | 0.226 | 28.942 | 0.857 | 0.174 |
| GKMNet [1] | NeurIPS 2021 | 25.468 | 0.789 | 0.219 | 29.081 | 0.867 | 0.171 |
| MDP [13] | WACV 2022 | 25.347 | 0.763 | 0.268 | 28.069 | 0.834 | 0.185 |
| DRBNet [3] | CVPR 2022 | 25.485 | 0.792 | 0.254 | 30.253 | 0.883 | 0.147 |
| MPRNet [16] | AAAI 2023 | 25.730 | 0.792 | 0.232 | - | - | - |
| Restormer [30] | CVPR 2022 | 25.980 | 0.811 | 0.178 | - | - | - |
| INIKNet [18] | ICCV 2023 | 26.055 | 0.803 | 0.185 | 30.293 | 0.886 | 0.132 |
| NRKNet [17] | CVPR 2023 | 26.109 | 0.810 | 0.210 | 30.481 | 0.884 | 0.147 |
| P2IKT [20] | AAAI 2024 | 26.288 | 0.807 | 0.191 | - | - | - |
| GGKMNet [19] | TPAMI 2024 | 26.272 | 0.810 | 0.215 | 30.552 | 0.886 | 0.154 |
| PPTFormer [31] | AAAI 2025 | 26.135 | 0.807 | 0.193 | - | - | - |
| DefocusGS [21] | ICCV 2025 | 26.431 | 0.827 | 0.171 | 30.660 | 0.889 | 0.129 |
| MP$^3$ (Ours) | - | 26.482 | 0.814 | 0.177 | 30.986 | 0.897 | 0.125 |

**Qualitative results.** We present visual comparisons with three representative open-sourced SOTA methods including NRKNet [17], INIKNet [18], and P2IKT [20] on RealDOF, CUHK-BD, and RTF in Fig. 3, using models trained on DPDD. Our method effectively suppresses halo artifacts and recovers finer details with fewer distortions in real-world scenarios.

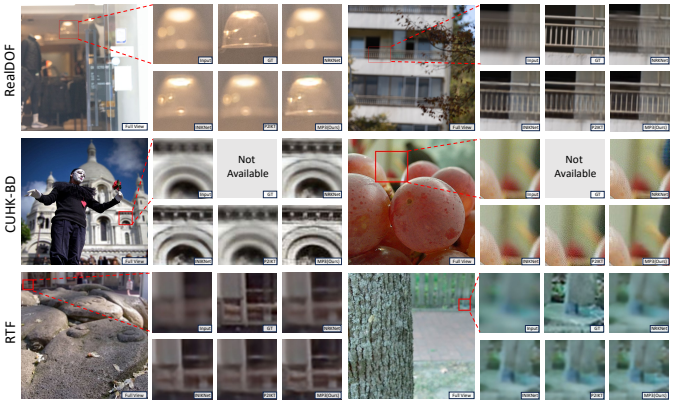



Fig. 3. Qualitative results on RealDOF (1st row), CUHK-BD (2nd row) and RTF (3rd row) with models trained on DPDD. Better zoom in for details.

### C. Ablation Studies and Analysis

**Quality metric.** Three variants are used to ablate the proposed metric: 1) Plain-sampling, which employs uniform random

TABLE II
QUANTITATIVE COMPARISONS ON REALDOF/RTF. BEST AND SECOND BEST ARE HIGHLIGHTED.

| | Model | Source | RealDOF PSNR | RealDOF SSIM | RealDOF LPIPS | RTF PSNR | RTF SSIM | RTF LPIPS |
|---|---|---|---|---|---|---|---|---|
| Trained on DPDD | DPDNet [28] | ECCV 2020 | 22.870 | 0.670 | 0.425 | 23.608 | 0.591 | 0.296 |
| | AIFNet [11] | TCI 2021 | 23.093 | 0.680 | 0.413 | 24.041 | 0.758 | 0.289 |
| | MDP [13] | WACV 2022 | 23.500 | 0.681 | 0.444 | 24.012 | 0.738 | 0.312 |
| | KPAC [15] | ICCV 2021 | 23.975 | 0.762 | 0.338 | 24.618 | 0.777 | 0.236 |
| | IFANet [2] | CVPR 2021 | 24.712 | 0.748 | 0.306 | 24.924 | 0.801 | 0.227 |
| | GKMNet [1] | NeurIPS 2021 | 24.254 | 0.732 | 0.392 | 24.970 | 0.789 | 0.261 |
| | DRBNet [3] | CVPR 2022 | 24.884 | 0.751 | 0.376 | 24.463 | 0.773 | 0.311 |
| | MPRNet [16] | AAAI 2023 | 24.541 | 0.736 | 0.339 | 24.588 | 0.788 | 0.304 |
| | Restormer [30] | CVPR 2022 | 25.091 | 0.762 | 0.285 | 24.212 | 0.821 | 0.224 |
| | NRKNet [17] | CVPR 2023 | 25.148 | 0.768 | 0.338 | 25.931 | 0.829 | 0.215 |
| | INIKNet [18] | ICCV 2023 | 25.231 | 0.765 | 0.287 | 25.450 | 0.834 | 0.215 |
| | GGKMNet [19] | TPAMI 2024 | 25.355 | 0.770 | 0.291 | 26.012 | 0.846 | 0.210 |
| | P2IKT [20] | AAAI 2024 | 25.481 | 0.762 | 0.306 | 25.857 | 0.839 | 0.207 |
| | DefocusGS [21] | ICCV 2025 | 25.567 | 0.772 | 0.281 | 26.114 | 0.855 | 0.207 |
| | MP$^3$ (Ours) | - | 25.650 | 0.777 | 0.275 | 26.734 | 0.886 | 0.118 |
| Trained on LFDOF | IFANet [2] | CVPR 2021 | 22.504 | 0.669 | 0.483 | 26.437 | 0.838 | 0.238 |
| | KPAC [15] | ICCV 2021 | 22.550 | 0.671 | 0.457 | 25.959 | 0.803 | 0.230 |
| | AIFNet [11] | TCI 2021 | 22.623 | 0.667 | 0.461 | 27.552 | 0.882 | 0.176 |
| | GKMNet [1] | NeurIPS 2021 | 23.609 | 0.721 | 0.408 | 26.985 | 0.856 | 0.246 |
| | MDP [13] | WACV 2022 | 22.726 | 0.680 | 0.453 | 25.580 | 0.809 | 0.228 |
| | DRBNet [3] | CVPR 2022 | 22.910 | 0.691 | 0.437 | 26.717 | 0.853 | 0.200 |
| | NRKNet [17] | CVPR 2023 | 23.528 | 0.716 | 0.431 | 28.047 | 0.889 | 0.145 |
| | INIKNet [18] | ICCV 2023 | 23.810 | 0.724 | 0.356 | 27.401 | 0.885 | 0.179 |
| | GGKMNet [19] | TPAMI 2024 | 24.108 | 0.735 | 0.385 | 28.308 | 0.905 | 0.140 |
| | DefocusGS [21] | ICCV 2025 | 24.219 | 0.750 | 0.362 | 28.451 | 0.909 | 0.136 |
| | MP$^3$ (Ours) | - | 24.706 | 0.753 | 0.326 | 28.481 | 0.914 | 0.075 |

patch sampling; 2) $\delta$-only, which considers only entropy-density decay; and 3) $\kappa$-only, which considers only residual sparsity. Quantitative results are shown in Tab. III. For Plain-sampling and $\delta$-only, while performance slightly drops on DPDD, generalization to RealDOF and RTF shows a significant decline. For $\kappa$-only, training favors patch pairs with highly sparse residuals, which weakens the learning signals, leading to a substantial performance drop.

**Analysis regarding $\delta$ and $\kappa$.** $\delta$ is highly related to richness of learning signals but cannot reflect purity. Thus $\kappa$ is jointly used for filtering out impure patch pairs. Fig. 4 illustrates visual comparisons between low-quality and high-quality patches. Within a single image, different regions show distinct pivot/purity. The top row patch shows dense residuals but a mismatched low entropy-density decay, while the bottom row shows both high $\delta$ and high $\kappa$ indicating the rich learning signal together with pure defocus degradation. Fig. 4 also illustrates that $\kappa$ for purity is physically grounded in the nature of defocus blur: since defocus kernels essentially act as low-pass filters and natural images are predominantly low-frequency, the resulting residuals are inherently sparse and primarily concentrated at edges. Conversely, dense residuals often signal non-physical artifacts, such as misalignment (as shown in Fig. 4), which violate the defocus prior. Consequently, $\kappa$ serves as a robust purity metric to identify reliable signals.

**Curriculum frequency learning.** To analyze the effect of curriculum strategy, under the same number of training steps and same frequency band range, we design three variants: 1) 1-stage: the non-curriculum version; 2) 2-stages: remove the middle stage; 3) 4-stages: add a new stage. As shown in Tab. III, without curriculum strategy, performance slightly drops across different datasets. Using more than 3 stages does not show further improvement and 3-stages training is enough.

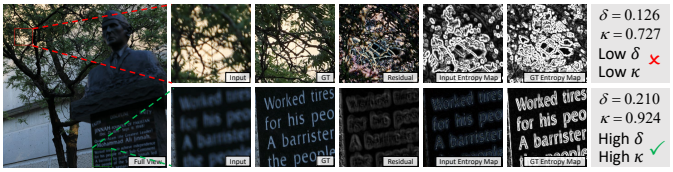



Fig. 4. Comparison between low-quality patches and high-quality ones.

TABLE III
QUANTITATIVE RESULTS OF ABLATION STUDIES.

| Method | DPDD PSNR | DPDD SSIM | DPDD LPIPS | RealDOF PSNR | RealDOF SSIM | RealDOF LPIPS | RTF PSNR | RTF SSIM | RTF LPIPS |
|---|---|---|---|---|---|---|---|---|---|
| Plain-sampling | 26.242 | 0.808 | 0.183 | 25.323 | 0.769 | 0.282 | 25.852 | 0.851 | 0.196 |
| $\delta$-only | 26.336 | 0.809 | 0.182 | 25.418 | 0.771 | 0.280 | 26.128 | 0.864 | 0.187 |
| $\kappa$-only | 26.065 | 0.804 | 0.189 | 25.179 | 0.763 | 0.289 | 25.946 | 0.857 | 0.192 |
| 1-stage | 26.329 | 0.810 | 0.184 | 25.483 | 0.771 | 0.279 | 26.454 | 0.871 | 0.129 |
| 2-stages | 26.423 | 0.811 | 0.181 | 25.547 | 0.774 | 0.277 | 26.596 | 0.878 | 0.133 |
| 4-stages | 26.430 | 0.813 | 0.180 | 25.607 | 0.776 | 0.276 | 26.757 | 0.888 | 0.113 |
| ConvLSTM | 26.457 | 0.814 | 0.174 | 25.597 | 0.776 | 0.274 | 26.677 | 0.884 | 0.120 |
| Direct-concat | 26.306 | 0.805 | 0.186 | 25.285 | 0.760 | 0.286 | 25.891 | 0.859 | 0.198 |
| Ours (full) | 26.482 | 0.814 | 0.177 | 25.650 | 0.777 | 0.275 | 26.734 | 0.886 | 0.118 |

**DNN architecture.** We construct two variants to analyze SCS module: 1) ConvLSTM: We ablate SCS with the widely-adopted ConvLSTM architecture; 2) Direct-concat: replace the selection with direct concatenation. The ConvLSTM variant shows comparable performance to ours but contains redundant activations and operations. In contrast, we preserve only the key cross-scale selections and achieve slightly better results. The Direct-concat variant shows importance of the selection mechanism.

**Analysis on cross-scale selections.** To better understand the effect of cross-scale selections, we visualize $\mathcal{A}^{(t)}$ in Fig. 5. Generally, we find that coarser scales concentrate on handling large defocus blur, whereas finer scales refine moderate blur. This observation suggests that the model adopts an economical strategy by assigning larger blur to coarser scales, which aligns with our design intuition.

## V. CONCLUSION AND FUTURE WORK

In this paper, we address the challenge of mining pure pivot patches essential for robust real-world single-image defocus deblurring. We propose an efficient quality metric that identifies patches with rich and clean supervision signals, leveraging two key observations: (i) large entropy-density decay indicates strong supervisory cues, and (ii) the residual between defocused and sharp counterparts must be highly sparse. By filtering low-quality patches and biasing training

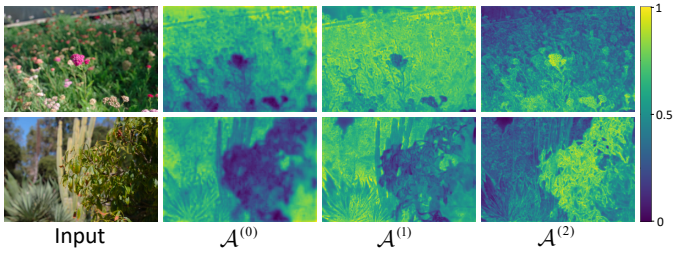



Fig. 5. Visualization of $\mathcal{A}^{(t)}$, obtained by averaging along the channel dimension.

toward high-quality ones, our approach yields notable generalization gains. We further introduce a lightweight network that eliminates redundant activations while retaining cross-scale selection, paired with a curriculum frequency learning strategy that progressively builds robustness from coarse structures to fine details.

Although promising, the current metric and its training integration remain preliminary, offering clear avenues for future refinement.

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